

Autism Spectrum Disorder

Presented by Angela Merrill!

Description and Diagnosis

- DSM-5:
 - Difficulties in social communication and social interaction
 - Restricted and repetitive behavior, interests, or activities
- Encompasses a wide **spectrum** of conditions and experiences
- Symptoms often confounded with **co-occurring conditions**
- Diagnosed with **questionnaires** and **observed activities**

Treatment

- Currently no effective drugs
- Behavioral Therapies
- Support/Accommodations
 - Free and Appropriate Public Education
- Tools for communication and emotional regulation

Behavioral Characterization

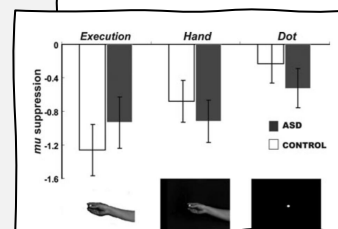
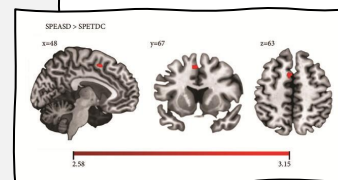
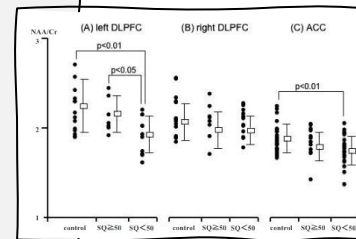
- Impaired Theory of Mind
 - Person Perception
 - Communication
- Executive Dysfunction
 - High-level: Need for predictability
 - Low-level: Impulses
- Altered sensitivity
- Hyperfixation, masking, stimming, meltdowns...

Impacts on the Brain

- Social Mind
 - Amygdala
 - Insula
- Mirror Neuron System
- Default Network
- Abnormal **Connectivity**
 - Corpus Callosum
 - Frontal Lobes
 - Long-range vs local

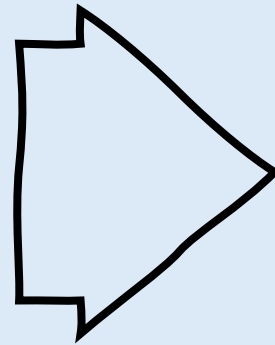
Research Summary

1. Fujii et al. 2009: Correlation between **brain metabolism in left DLPFC and ACC, social quotient, and autism**
2. Mosner et al. 2019: **Higher activation** in paracingulate gyrus and anterior insula during **prediction errors**.
3. Fan et al. 2010: **Mirror Neuron System** is disrupted during **observation of intransitive movements**



Description^[4]

- **Spectrum Disorder**
 - Asperger's Syndrome
 - Autism
 - Childhood Disintegrative Disorder
 - Pervasive Developmental Disorder-Not Otherwise Specified (PDD-NOS)
 - 5 Levels (Support Needs)
- Traditionally diagnosed in **young boys**
- **Causes**
 - Genetics
 - Pregnancy complications
 - Gestational exposure to chemicals
 - Abnormal stem cell proliferation



Multiple impacts across the brain

Diagnosis^[4]

- DSM-5:
 - Difficulties in social communication and social interaction
 - Restricted and repetitive behavior, interests, or activities
- Typical co-occurring conditions:
 - Intellectual Disability
 - Language Disorders
 - ADHD
 - Tics
 - Motor Abnormality
 - Epilepsy
 - Immune dysregulation
 - Sleep disorders
 - Anxiety
 - Depression
 - OCD
 - Psychotic Disorders
 - Personality Disorders
 - Pica
 - Auditory Processing Disorder

Checklist for Autism in Toddlers (CHAT)^[B]

Does your child....

1. Enjoy being swung/bounced on your knee/etc.?
2. Take an interest in other children?
3. Like climbing on things, such as stairs?
4. Enjoy playing peek-a-boo/hide-and-seek?
5. **Ever play pretend, like taking care of dolls?**
6. Use their index finger to point, to ask for something?
7. **Use their index finger to point, to indicate interest in something?**
8. Play properly with small toys without just mouthing/fiddling with/dropping them?
9. Bring objects to you to show you something?

Behavioral Characterization^[4]

Restricted/
Repetitive
Actions

Difficulties in
Socialization

Difficulty
Combining/
Splitting Steps

Executive
Dysfunction

Impaired Theory
of Mind

Intolerance of
Change

Hypo-/
Hypersensitivity

Impaired
Top-Down &
Bottom-Up

Impaired Person
Perception

Increased
Pattern
Recognition

Behavioral Characterization^[4]

Restricted/
Repetitive
Actions

Difficulties in
Socialization

Systemizing

Perfectionism

Deficits in
Working
Memory

Impaired
Mentalizing

Hyperfixation

Difficulty
Combining/
Splitting Steps

Lack of
Mirroring
Others

Executive
Dysfunction

Impaired Theory
of Mind

Intolerance of
Change

Difficulty
Suppressing
Impulses

Masking

Literal
Thinking

Special
Interests

Stimming

Hypo-/
Hypersensitivity

Impaired
Top-Down &
Bottom-Up

Difficulty
Reading
Faces

Impaired Person
Perception

Increased
Pattern
Recognition

Focus on
Details

Meltdowns

Reduced Eye
Contact

Lack of
Interoception

Black-and-
White Thinking

Impacts on the Brain

Increased
Activity

Abnormal

Decreased
Activity

Social Brain^[4, 5]

Medial PFC

Inferring

Amygdala

Fusiform Gyrus

Tempoparietal
Junction

Reading
Faces /
Mentalizing

Inferior
Frontal Gyrus

Posterior STS

Eye Contact

Anterior Insula

Imitation

Inferior
Parietal Sulcus

Spatial
Attention

Mirror Neuron System^[3, A, B]

Action Chains

Firing Rates

Default Network^[A]

Medial PFC

Tempoparietal
Junction

Ventral ACC

Interoception

Recruitment/Connectivity^[5]

Local Connectivity

Long-Distance Connectivity

Corpus Callosum

Synchronization of
Language Areas

Anterior and Posterior
Cortex Connectivity

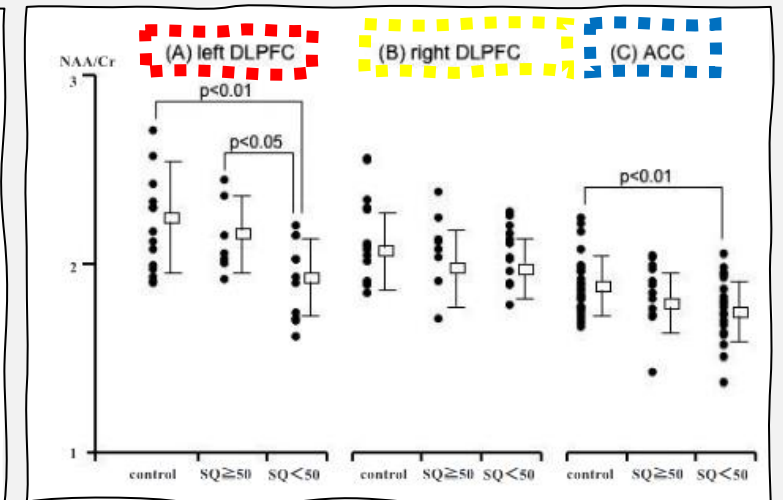
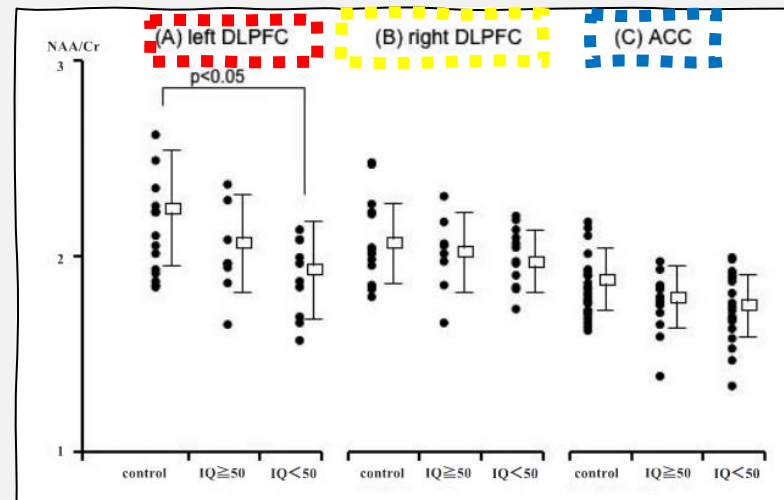
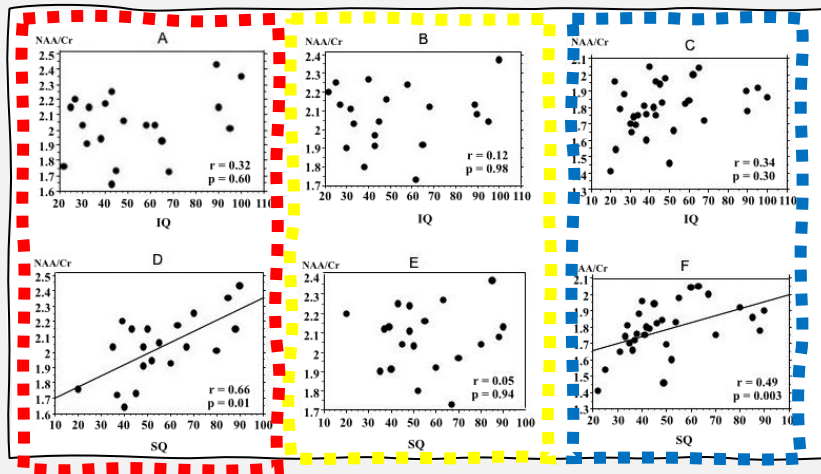
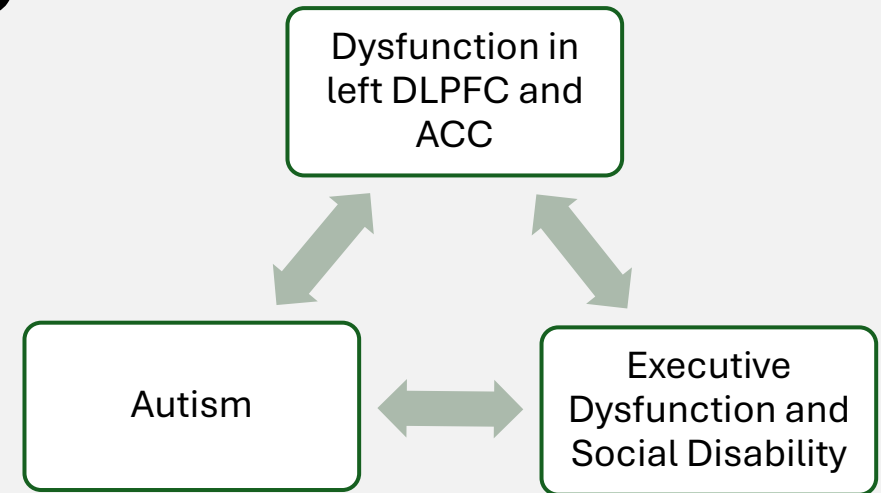
Associative and Frontal
Cortex Connectivity

Primary Sensory Cortexes

Synchronization of
Parietal and Occipital

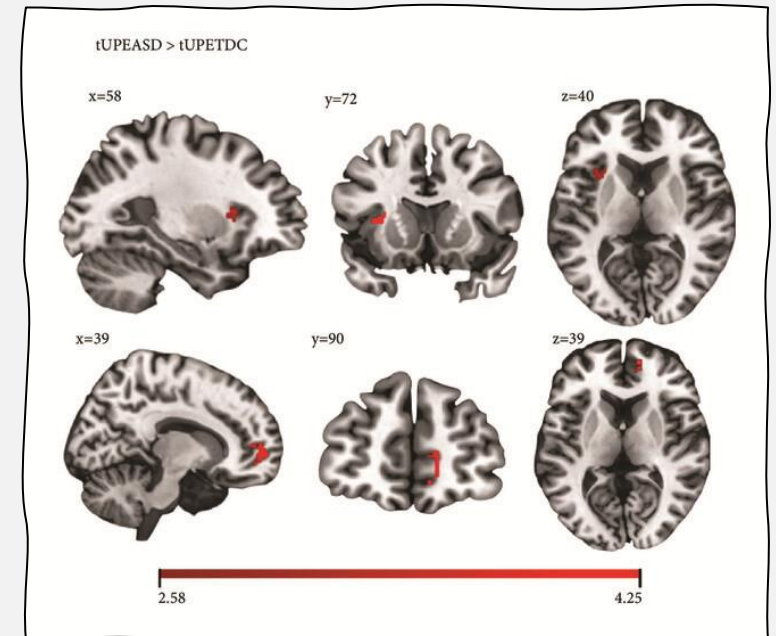
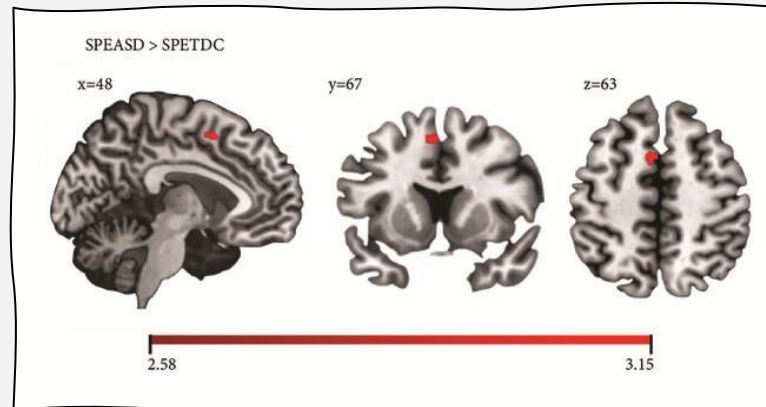
Research Study #1: Fujii et al. 2009

- Proton Magnetic Resonance Spectroscopy
 - Assess brain metabolism/cell density
 - NAA/Cr in left/right DLPFC and ACC
- Significantly lower NAA/Cr in:
 - Left DLPFC for IQ<50
 - Left DLPFC and AAC for SQ<50



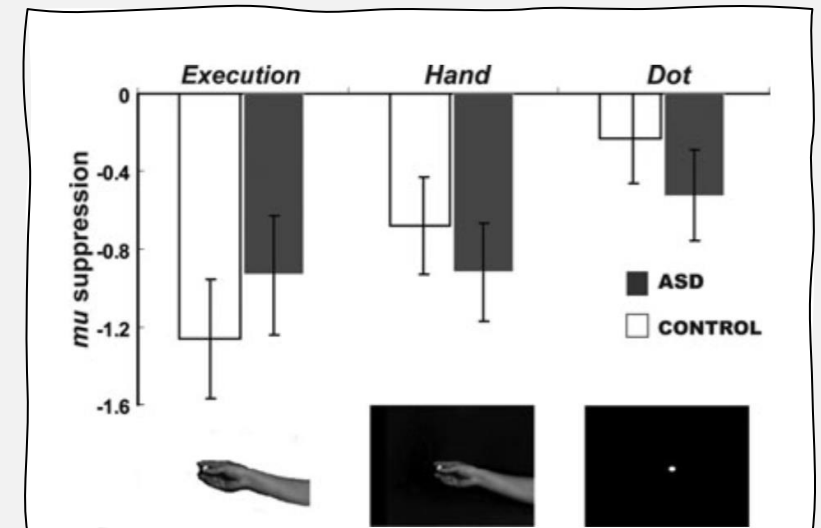
Research Study #2: Mosner et al. 2019

- Reward Prediction Error Task with fMRI
- Task: select the cue (quilt square) associated with a monetary reward
 - Cues become less consistent
- Greater activation in autistic patients:
 - Unexpected Reward vs Nonreward: Paracingulate gyrus
 - Expected vs Unexpected Rewards: Left anterior insula and right frontal pole
- Explains higher sensitivity to unpredictability
 - Social situations
 - Theory of Mind



Research Study #3: Fan et al. 2010

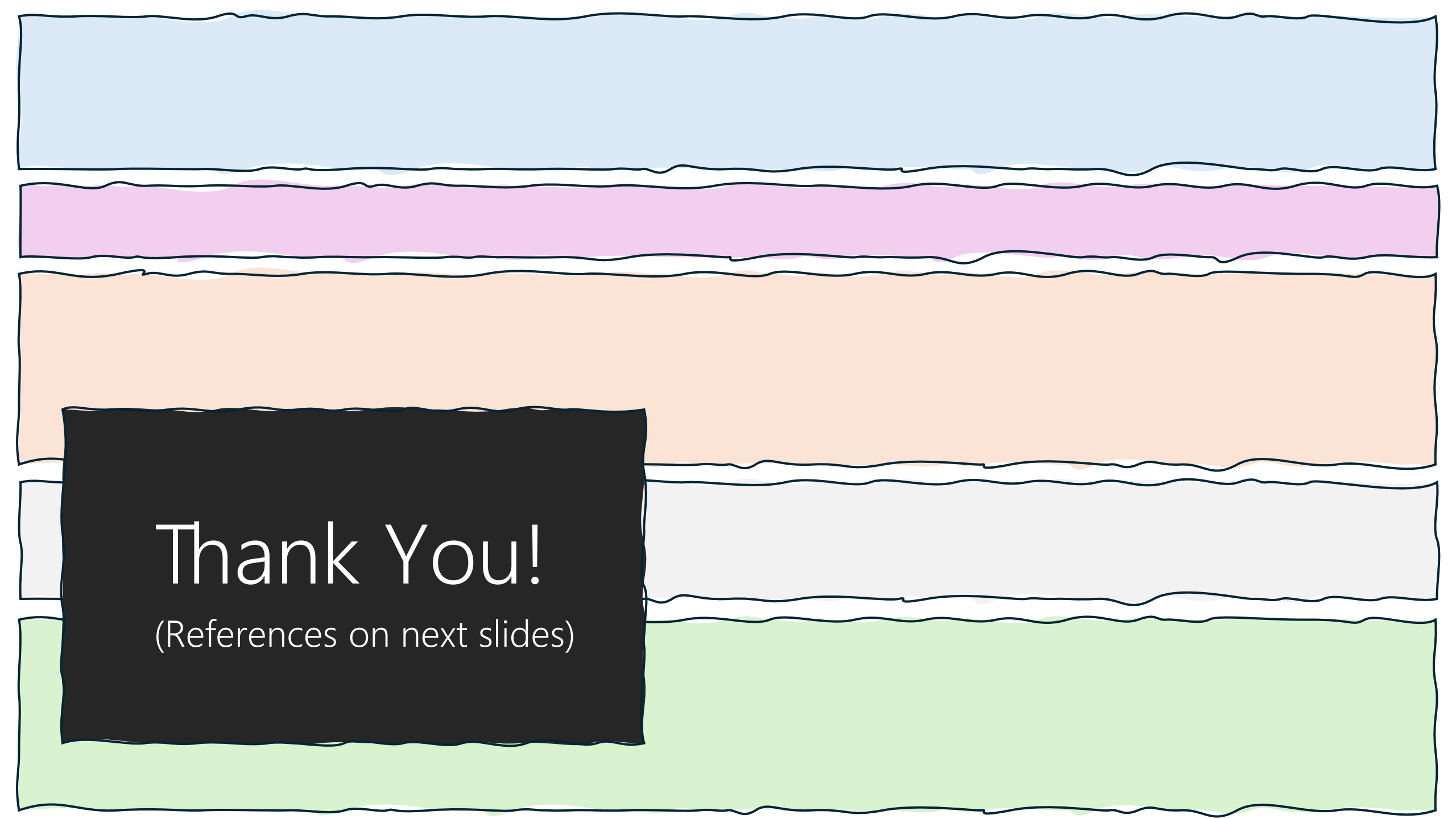
- Mirror Neuron System (MNS) and EEG mu waves
- Tasks:
 - Video of moving dot - control for movement
 - Video of a hand manipulating an object - observation
 - Moving own hand - execution
- In neurotypical humans, mirror neurons = socialization^[C]
 - Imitation
 - Language
 - Theory of Mind
 - Empathy
- Evidence that MNS is not completely disrupted
 - Previous data: MNS is disrupted in observation^[C]
 - New data: MNS is disrupted in **observing intransitive actions**



Treatment^[4, C]

- As a neurotype, there will never be a “cure”
- Potentially Effective Drugs
 - Antipsychotics – reduce repetitive behaviors
 - SSRIs – reduce repetitive behaviors
 - ADHD stimulants – treat co-occurring ADHD symptoms
- Therapy (Speech, Occupational, DBT, etc.)
- Free and Appropriate Public Education^[E] (“Special Needs” classes)
- Behavioral Approaches
 - Discrete Trial Training (cue → action → reward)
 - Pivotal Response Treatment (rewarding attempts at social interactions)
- Tools
 - Picture Exchanged Communication System
 - Noise-cancelling headphones
 - Visual instructions





Thank You!

(References on next slides)

References

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